($N/m^2$). In physics, the wavelength of a sinusoidal wave is the spatial period of the wave—the distance over which the wave's shape repeats. The unit for wavelength is angstrom. In physics, energy is a quantity that is the ability to do work. In the International System of Units (SI), energy is measured in joules, but in some fields other units such as kilowatt-hours and kilocalories are also used.

**103. For which one of the following is capillarity not the only reason?**

(a) Blotting of ink
(b) Rising of underground water
(c) Spread of water drop on a cotton cloth
(d) Rising of water from the roots of a plant to its foliage

**Ans. (b) :** Capillary action, or capillarity, is a phenomenon where liquid spontaneously rises in a narrow space such as a thin tube, or in porous materials such as paper or in some non-porous materials such as liquefied carbon fibre. This effect can cause liquids to flow against the force of gravity or the magnetic field induction. In blotting of ink, spread of water drop on a cotton cloth and the rising of water from the roots of a plant to its foliage etc. are example of it whereas for rising of underground water capillarity is not the only reason. It can also rise due to water recharge also.

**104. Consider the following statements:**

**1. If a person looks at a coin which is in a bucket of water, the coin will appear to be closer than it really is.**
**2. If a person under water looks at a coin above the water surface, the coin will appear to be at a higher level than it really is.**

**Which of the above statements is/are correct?**

(a) 1 and 2 (b) 1 alone
(c) 2 alone (d) Neither 1 nor 2

**Ans. (a) :** When the person looks at a coin which is in a bucket of water, the coin will appear closer and at a higher level than it really is. μ2 = real depth / apparent depth.

## CHEMISTRY

**105. When ants bite, they inject**

(a) Glacial acetic acid (b) Methanol
(c) Formic acid (d) Satiric acid

**Ans. (c) :** When ants bite, they inject formic acid in the muscle tissue. This formic acid is found at the mandible of ants.

**106. Consider the following statements:**
**Glass can be etched or scratched by**
**1. Diamond 2. Hydrofluoric acid**
**3. aquaregia 4. conc. sulphuric acid**
**Which of these statements are correct?**

(a) 1 and 4 (b) 2 and 3
(c) 1 and 2 (d) 2 and 4

**Ans. (c) :** Glass is etched by applying a caustic compound to the glass surface. The compound chemically deposits at the glass substrate (borosilicate), and causes etching in the glass surface. Another method involves physically grinding the glass surface with an abrasive, like a grinding wheel. The surface is worn away where the wheel hits, and the etched surface remains. Some very artistic designs can be created with both methods. Glass can be etched or scratched by diamond and hydrofluoric acid.

**107. Assertion (A) : To dilute sulphuric acid, acid is added to water and not water to acid.**
**Reason (R) : Specific heat of water is quite large.**

(a) Both A and R are True and R is the correct explanation of A
(b) Both A and R are True but R is not a correct explanation of A
(c) A is true but R is False
(d) A is false but R is True

**Ans. (a) :** Sulphuric acid (H2SO4) reacts very vigorously with water, in a highly exothermic reaction. Thus if you add water to concentrated sulfuric acid, it can boil and you may get a nasty acid burn. That's why for dilution, acid is added to water not water to acid as specific heat of water is quite large and it can absorb large quantity of heat produced by sulphuric acid. Moreover water is less dense than sulphuric acid, so if water is poured on the acid, the reaction occurs on top of the liquid. If acid is added to the water, it sinks and any wild and crazy reactions have to get through the water or beaker.

**108. Match List-I (Naturally occurring substances) with List-II (Elements) and select the correct answer using the codes given below the lists:**

| List-I | List-II |
|---|---|
| A. Diamond | 1. Calcium |
| B. Marble | 2. Silicon |
| C. Sand | 3. Aluminium |
| D. Ruby | 4. Carbon |

**Codes:**

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 3 | 1 | 2 | 4 |
| (b) | 4 | 2 | 1 | 3 |
| (c) | 2 | 1 | 3 | 4 |
| (d) | 4 | 1 | 2 | 3 |

**Ans. (d) :** Diamonds is the polymorph of the element carbon. Calcium is the basic element of naturally occurring marble. Sand is formed by Silicon and Aluminium is the basic element of naturally occurring Ruby.



## BIOLOGY

**109. Assertion (A) : Insect resistant transgenic cotton has been produced by inserting Bt gene.**
**Reason (R) : The Bt gene is derived from a bacterium.**

(a) Both A and R are True and R is the correct explanation of A
(b) Both A and R are True but R is not a correct explanation of A
(c) A is true but R is False
(d) A is false but R is True

**Ans. (a) :** Bacillus thuringiensis is the bacterium which is used in Bt gene of cotton. Bt cotton is insect resistant.

**110. Assertion (A) : Dolly was the first cloned mammal.**
**Reason (R) : Dolly was produced by in vitro fertilization.**

(a) Both A and R are True and R is the correct explanation of A
(b) Both A and R are True but R is not a correct explanation of A
(c) A is true but R is False
(d) A is false but R is True

**Ans. (d) :** Cloning is a process, which is artificially performed in laboratory ( in vitro). Dolly was a cloned animal, but not the first cloned mammal. In 1984, a mice was cloned first.

**111. In eye donation, which one of the following parts of donor's eye is utilized?**

(a) Iris (b) Lens
(c) Cornea (d) Retina

**Ans. (c) :** Cornea, the main focusing part , is the clear front surface of the eye. Like a window, it allows light to enter the eye. Vision could be partially reduced or lost if the cornea becomes cloudy or scarred. This condition is known as corneal blindness. Eye donation is an act of donating one's eyes after his/her death. Only corneal blinds can be benefited through this process not other blinds. Cornea transplant is the surgical procedure which replaces a disc-shaped segment of an impaired cornea with a similarly shaped piece of a healthy donor cornea.

**112. Match the hormones in List-I with items in List-II and select the correct answer using the codes given below the lists:**

| List-I | List-II |
|---|---|
| A. Adrenalin | 1. Anger, fear, danger |
| B. Estrogen | 2. Attracting partners through sense of smell |
| C. Insulin | 3. Females |
| D. Pheromones | 4. Glucose |

**Codes:**

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 3 | 1 | 4 | 2 |
| (b) | 1 | 3 | 2 | 4 |
| (c) | 1 | 3 | 4 | 2 |
| (d) | 3 | 1 | 2 | 4 |

**Ans. (c) :** Adrenalines are secreted in the situations of exercise, fear or any kind of dangerous situation. Oestrogens are steroid hormones or female primary sex hormones. Insulin is responsible for the metabolism of sugar in the body. Pheromones are the compounds which are secreted generally from insect. Some of the pheromones include alarm pheromones, food trail pheromones, and sex pheromones.

**113. Neem tree has acquired industrial importance as a source of:**

(a) Biopesticide and antifertility compound
(b) Antifertility compound, biofertilizer and anti-cancer drug
(c) Biofertilizer, biopesticide and antifertility compound
(d) Anti-cancer drug, biopesticide and biofertilizer

**Ans. (c) :** Neem has been patented by India for its use as biofertilizer, biopesticide and antifertility compound as contraceptive. Neem is a natural herb that comes from the neem tree, other names for which include Azadirachta indica and Indian lilac. The extract comes from the seeds of the tree and has many different traditional uses. Neem is known for its pesticidal and insecticidal properties, but people also use it in hair and dental products

**114. Match List-I (Drugs/Chemicals) with List-II (Their uses) and select the correct answer using the codes given below the lists:**

| List-I | List-II |
|---|---|
| A. Atropine | 1. Local anesthesia |
| B. Ether | 2. Heart trouble |
| C. Nitroglycerine | 3. Dilation of pupil |
| D. Pyrethrin | 4. Mosquito control |

**Codes:**

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 1 | 3 | 2 | 4 |
| (b) | 1 | 3 | 4 | 2 |
| (c) | 3 | 1 | 4 | 2 |
| (d) | 3 | 1 | 2 | 4 |

**Ans. (d) :** Ether is used as an anesthesia at the time of surgery to prevent pain. Atropine is used as eye drop to dilate the pupil before eye examination. Nitroglycerine is widely used in the treatment of heart. Pyrethrin is biodegradable, non-persistent pesticide which is used to kill mosquitoes, fleas and other insects.